The Physical Substrate of the Soma-Field

Biotensegrity, Fascial Interoception, and Bioelectric Correlates of Emotional Field Dynamics

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Abstract

The Soma-Field model describes emotional dynamics as a tensor-valued field over an attractor landscape, formally equivalent to a Hopfield network with topological structure (Johnson, 2026c). The model is mathematically complete: it derives its dynamics from first principles, generates falsifiable predictions, and has passed pre-registered computational validation (QUANT-EXP-1) (Johnson, 2026b). What it does not address is the physical substrate in which such a field is instantiated in living tissue.

This paper identifies three bodies of empirical research that jointly constitute that substrate. Biotensegrity (Ingber, 1997, 2003; Levin, 2002) establishes that the body is a pre-tensioned continuous mechanical network at all scales: this is the architecture through which the somatic wave propagates globally rather than locally. Fascial-interstitial continuity research (Langevin & Huijing, 2009; Oschman, 2016; Schleip, 2003) identifies fascia as an active innervated signalling tissue — the physical pathway of interoception — and documents that chronic trauma produces measurable increases in fascial stiffness that correspond quantitatively to deep Hopfield attractor basins. Biofield physiology (Ho, 1998; McCraty & Childre, 2010; Popp, 2003; Rubik et al., 2015) documents coherent electromagnetic and biophotonic emissions from living tissue that are the most plausible physical candidates for the field correlate itself.

Three explicit bridges are developed. Fascial armoring corresponds to high energy barriers in the attractor landscape; myofascial release is barrier lowering rather than barrier crossing; therapist-client physiological entrainment is the physical mechanism of mathematical co-identification (Johnson, 2026a). Each bridge generates testable predictions outside the computational domain.

Contents

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1 The Missing Layer

The Soma-Field model (Johnson, 2026c) establishes that the limbic system and its somatic coupling are governed by the same formal apparatus as a quantum field on a manifold: tensor-valued dynamics, Hopfield energy functionals, topological barriers between attractor states. The identification is not an analogy; it is a co-identification in the technical sense (Johnson, 2026a) — the governing equations are the same equations, and every theorem of the source domain imports into the target.

That mathematical work is complete. What it leaves open is a question that sits one level below the mathematics: *what is the body made of, such that it could host a field like this?*

The Hopfield attractor landscape is an abstract object. For it to describe a physical organism, there must be a physical substrate — tissue, architecture, medium — that implements the attractor dynamics, propagates the somatic wave, and generates the coherent state that the mathematics describes. The model says there is a somatic wave $\mathbf{E}_{\text{body}}(x,t)$. The question is: what physical thing is that wave a description of?

Three independent bodies of experimental and theoretical research converge on this substrate. They were developed largely in parallel, each with its own language, none formulated with the soma-field model in mind. This paper argues they are describing the same system at three different scales of resolution:

1. **Architecture** (§2): Biotensegrity theory establishes the mechanical network structure through which somatic signals propagate globally. This is the physical basis for the spatial extent of $\mathbf{E}_{\text{body}}(x,t)$.

2. **Substrate** (§3): Fascial-interstitial continuity research identifies the specific tissue — fascia — that constitutes the body-wide signalling medium, and documents the interoceptive pathway from peripheral tissue to cortical representation. The quantitative correspondence between fascial stiffness and attractor depth is developed here.

3. **Field correlate** (§4): Biofield physiology documents coherent electromagnetic and biophotonic emissions from living tissue that are the most plausible physical candidates for the field itself — not the network that hosts it, but the coherent state that the network generates.

Section §5 states the three explicit bridges to the formal model. Section §6 lists the testable predictions that follow.

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2 Biotensegrity: The Architecture of the Somatic Wave

2.1 The Lever Model is Wrong

Standard physiological and biomechanical models treat the body as a rigid-lever system: bones as struts, muscles as cables pulling across pin-joint connections, forces transmitted locally from joint to joint. This model works tolerably well for gross locomotion analysis but fails to account for whole-body responses to local perturbation, and fails completely at the cellular scale.

Ingber's tensegrity model (Ingber, 1997, 2003) replaces this with a different architecture. In a tensegrity structure, rigid compression elements (in the body: bone, cartilage) float within a continuous network of tension elements (fascia, tendon, ligament, muscle), and mechanical prestress is distributed throughout the network simultaneously. There are no isolated pin-joints: the whole system is pre-loaded, so perturbation anywhere propagates everywhere.

Levin extended this framework to the full organism (Levin, 2002), arguing that biotensegrity is not merely a useful approximation but the correct architectural description at every scale: from the cytoskeleton of individual cells through the deep fascial planes to gross musculoskeletal anatomy. Each scale implements the same tensegrity geometry. Each is mechanically continuous with the others. The architecture is fractal.

2.2 Global Propagation

The clinical consequence of this architecture is direct: mechanical information does not travel locally through joint-to-joint lever chains. It propagates through the prestressed fascial network to the whole organism simultaneously, with the spatial distribution governed by the topology and stiffness of the network rather than anatomical lever arms.

This is experimentally documented. Langevin's group showed that needle insertion at acupuncture points produces tissue displacement patterns propagating along fascial planes far from the insertion point, following biotensegrity-predicted paths rather than nerve or muscle routes (Langevin & Huijing, 2009). The body responds as a continuous tensioned whole, not as a collection of local structures.

The speed of this propagation is also relevant. Neural conduction (axonal) operates in milliseconds. Mechanical wave propagation through a prestressed medium operates in microseconds. For fast somatic responses — the startle reflex, the breath-hold, the full-body freeze — the biotensegrity medium is faster than the nervous system and spatially global in a way that the nervous system, with its point-to-point wiring, is not.

2.3 Correspondence to the Somatic Wave

The Soma-Field model posits a somatic wave $\mathbf{E}_{\text{body}}(x,t)$ — a field defined over the body, propagating continuously, carrying emotional-somatic information. The question "how can a perturbation in one body region give rise to a global somatic state?" has typically been answered by appeal to the nervous system: proprioception, interoception, vagal signalling. These are real and important, but they are axonal (slow, discrete) and do not account for the observed speed and spatial coherence of whole-body somatic responses.

Biotensegrity provides the continuous mechanical medium the model requires. The formal correspondence is:

> The body's biotensegrity network is the **physical implementation** of $\mathbf{E}_{\text{body}}(x,t)$. The tensor field over the body in the mathematical model corresponds to the mechanical stress tensor distributed across the fascial network in the physical organism.

Both are spatially extended. Both propagate continuously. Both couple to the neural wave at every point: every mechanoreceptor in the fascia is a coupling node between $\mathbf{E}_{\text{body}}$ and $\mathbf{E}_{\text{neural}}$.

The somatic wave is not *like* a wave in a continuous medium. In the fascial network, it *is* a wave in a continuous medium. The co-identification (Johnson, 2026a) is architectural.

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3 Fascial-Interstitial Continuity: The Pathway and the Armoring

3.1 Fascia as Active Signalling Tissue

The classical anatomical view of fascia — as inert white packaging, the sheaths that dissectors clear away to reach the "real" anatomy — was overturned by experimental work beginning in the 1990s.

Schleip's review (Schleip, 2003) documented that fascia contains all four classes of mechanosensory nerve endings: Ruffini corpuscles (respond to lateral stretch), Golgi tendon organs (respond to compression), Pacinian corpuscles (vibration and rapid changes), and type IV free nerve endings (polymodal: mechanical deformation, temperature, chemical changes). The type IV endings are especially significant: they project primarily to the insular cortex via lamina I of the spinal cord — the Craig interoceptive pathway (Craig, 2003) — and constitute the neurological substrate of body-felt emotional experience, not merely visceral sensation.

Langevin's work established that fascia actively participates in signalling: mechanical deformation produces fibroblast shape changes, cytoskeletal reorganisation, and gene expression changes on timescales from seconds to minutes (Langevin & Huijing, 2009). The tissue is a transducer, not a cable.

Oschman's "living matrix" model (Oschman, 2016) extends this further: the entire connective tissue system — fascia, interstitium, extracellular matrix — constitutes a continuous liquid-crystalline semiconductor network. Piezoelectric effects in collagen generate electrical potentials under mechanical stress. DC currents flow through the network continuously. The fascial system is simultaneously mechanical, chemical, and electrical.

3.2 The Interoceptive Pathway

Interoception — the body's sensing of its own internal state — is the somatic input channel of the Soma-Field model. It is the mechanism by which the body schema is updated and by which the energy functional of the attractor landscape is computed.

The fascial pathway of interoception is now well characterised (Craig, 2003; Garfinkel et al., 2016; Schleip, 2003): type IV free nerve endings in deep fascia, visceral fascia, and interstitial tissue → lamina I neurons of the dorsal horn → thalamus → anterior insular cortex. This is the Craig pathway, increasingly recognised as the neurological substrate of emotional experience proper, distinct from and complementary to the classical somatosensory pathway.

The clinical implication is direct: interoceptive dysfunction (well documented in ASC, CPTSD, and related conditions) (Garfinkel et al., 2016) is dysfunction of the fascial-to-insular projection. It is not merely a processing deficit in higher cortical areas; it originates in the tissue. Restoring interoceptive accuracy therefore requires working at the fascial level — which is precisely what somatic therapies (Somatic Experiencing, Sensorimotor Psychotherapy, EMDR somatic protocols, myofascial release) do, whether or not they are theorised in those terms.

3.3 Fascial Armoring as Attractor Depth

This section develops the most important connection in this paper.

Wilhelm Reich introduced "character armoring" — the clinical observation that chronic emotional states (fear, shame, traumatic holding) produce corresponding patterns of chronic muscular and somatic tension. The observation was clinically compelling but had no formal model. It was a phenomenology without a mechanism.

Schleip and subsequent workers (Stecco, Bordoni, Bhatt) documented the fascial component: chronic trauma produces not merely chronic muscular contraction but *measurable changes in fascial stiffness*, quantifiable by ultrasound elastography (Schleip, 2003). High-trauma individuals show significantly elevated fascial stiffness in characteristic body regions, with the spatial pattern reflecting the specific trauma history. The psoas, diaphragm, and posterior cervical chain are typically implicated in chronic fear responses; the pericardium and thoracic fascia in grief and heartbreak; the pelvic floor in sexual trauma. These are not metaphors. They are measured tissue properties.

In the Hopfield model, the attractor landscape is characterised by energy barriers W_{ij} between attractor states. The Fear basin has a high energy barrier. The computational experiment QUANT-EXP-1 (Johnson, 2026b) shows that cold classical dynamics cannot cross a barrier of $W = -8$ to $W = -14$.

The bridge:

$$\text{fascial stiffness at region } r \;\leftrightarrow\; |W_{ij}| \text{ for state transition involving } r$$

High fascial stiffness = high energy barrier. The organism is mechanically locked into the Fear attractor not only neurologically but anatomically — the tissue itself has been remodelled to implement the barrier. This is why van der Kolk's title (Kolk, 2014) is accurate in a way he could not have fully formalised: the body does not merely *express* the trauma; it *encodes* the attractor depth in its mechanical structure.

The quantitative claim is: the QUANT-EXP-1 barriers $W = -8, -10, -12, -14$ have physical correlates in fascial stiffness values measurable in kPa (shear wave elastography units). The mapping is not known yet — establishing it is part of the empirical programme in §6 — but the existence of the correspondence is now claimed by this paper.

3.4 Myofascial Release as Barrier Lowering

QUANT-EXP-1 demonstrates that quantum annealing can cross barriers that classical cold dynamics cannot. This was framed computationally. The fascial literature provides a physical translation that clarifies an important distinction.

Classical barrier crossing (hot classical or quantum): The system transitions from one attractor to another while the barrier remains intact. This corresponds to either high-arousal state transitions (classical thermal, i.e. highly activated emotional states) or the quantum mechanism identified in QUANT-EXP-1.

Myofascial release (barrier reduction): Manual intervention directly reduces fascial stiffness — measured pre/post by elastography. This does not push the system over the barrier. It *lowers* the barrier so that transitions become accessible by classical means.

This distinction — barrier lowering versus barrier crossing — may explain the phenomenology of different therapeutic modalities and why they are experienced differently:

- Somatic bodywork (Rolfing, myofascial release, craniosacral) *reshapes the landscape.* The client often reports gradual deepening ease, reduction of chronic holding, and access to emotional material that was previously unavailable without drama.
- High-intensity interventions (EMDR reprocessing, breathwork, certain trauma protocols) may be *crossing a barrier that remains intact*: sudden, non-linear transitions, sometimes dramatic releases, the characteristic "before and after" quality of a topological transition.

Both produce movement in the attractor landscape. The mechanism is different. The soma-field model, grounded in the fascial correspondence, now predicts this difference and makes it testable (§6).

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4 Biofield Physiology: The Field Correlate

4.1 Living Systems Emit Coherent Fields

The soma-field is a mathematical field — an abstract object defined by its equations. For it to be physically real rather than merely useful, it must have a physical correlate: some measurable property of the organism that corresponds to the field's state. Three lines of evidence point toward coherent electromagnetic and biophotonic emissions as candidates.

Biophoton emission (Popp, 2003): All living cells emit ultra-weak light in the visible to near-UV range. This is not blackbody radiation (which would require far higher temperatures) but coherent emission with photon statistics more consistent with laser emission than thermal sources. Popp argues that this biophotonic field constitutes a real-time communication channel, carrying coherent information across tissue faster than any biochemical signal. The field is state-sensitive: stress, illness, and emotional perturbation all produce measurable changes in biophotonic emission patterns.

Liquid crystalline living matrix (Ho, 1998): Ho's model proposes that the connective tissue system — specifically the liquid crystalline ordering of collagen, water, and proteoglycans — constitutes a quantum coherent medium. Proton conduction and electronic charge delocalisation through this medium produce a macroscopic coherent quantum state distributed across the organism. This is not the Penrose-Hameroff proposal (which is neuron-centred and operates via microtubules); Ho's coherent organism is body-centred, connective tissue-centred, and is precisely the medium in which the Soma-Field would propagate as a physical entity.

Heart-brain coherence (McCraty & Childre, 2010): The heart generates a toroidal electromagnetic field measurable at distances from the body, with spectral content reflecting the organism's emotional state. Heart rate variability (HRV) in the low-frequency band (approximately 0.1 Hz) indexes the balance between sympathetic and parasympathetic regulation — the physiological correlate of transitions between Fear-dominant and Awe-dominant states in the soma-field model. McCraty's group demonstrates that this field entrains between proximate individuals: measurable cardiac coherence synchronisation occurs between therapist and client, between individuals in rapport, and between individuals and coherent social environments. This entrainment is not inferred; it is measured by simultaneous ECG recording.

4.2 The Rubik Synthesis

Rubik, Muehsam, Hammerschlag, and Jain (Rubik et al., 2015) published a systematic review of the biofield hypothesis in 2015, collating evidence from biophoton research, bioelectromagnetics, traditional medicine, and clinical trials. Their conclusion is deliberately conservative: the biofield hypothesis — that living organisms generate and respond to coherent electromagnetic and biophotonic fields beyond what is explained by classical biochemistry — is supported by a substantial and growing body of evidence, but mechanism and theoretical framework remain contested.

From the Soma-Field perspective, the contest is tractable: the theoretical framework is the quantum field on a Hopfield attractor landscape, and the biofield is the physical manifestation of that field. The soma-field model does not prove the biofield; it provides the theoretical frame within which the biofield evidence becomes interpretable rather than anomalous.

What the soma-field predicts is that the biofield — whatever its physical implementation — will show attractor-like behaviour: it will tend to occupy characteristic states, resist perturbation away from those states, and show non-classical transitions between states when the barrier is sufficiently large. HRV coherence, biophotonic emission, and DC skin conductance are all candidate observables. Which observable best couples to which component of the tensor field is an empirical question that this model now makes precise enough to ask.

4.3 Scope and Epistemic Status

The biofield section of this argument carries more epistemic weight than §§2–3, and this should be stated explicitly. Biotensegrity and fascial signalling are well-supported by peer-reviewed experimental evidence in mainstream biomechanics, cell biology, and clinical science. The biofield claims are supported by evidence in a more contested terrain, and the proposed identification between the soma-field's formal structure and the organism's EM/biophotonic emissions is a hypothesis, not an established result.

What this paper claims in §4 is modest: these are the best current physical candidates for the field correlate; they are consistent with the formal model; they generate testable predictions (§6). The stronger claim — that the soma-field *is* the biofield, formally — requires empirical work that does not yet exist.

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5 Three Bridges to the Formal Model

This section states the three principal connections between the physical substrate literature and the formal soma-field model, in a form that makes their testability explicit.

5.1 Bridge 1: Fascial Armoring = Attractor Depth

Physical claim (Schleip, 2003): Chronic trauma produces chronically elevated fascial stiffness, measurable by ultrasound shear-wave elastography, with characteristic spatial patterns reflecting trauma type and history.

Formal correspondence: Fascial stiffness at region r maps to $|W_{ij}|$, the energy barrier between attractor states i and j in the Hopfield network, where r is the somatic representation zone of the relevant emotional state pair. High stiffness = high barrier = deep attractor basin.

Testable prediction (P1): Populations with documented high-barrier emotional states (CPTSD, complex trauma, chronic anxiety disorder) should show systematically elevated fascial stiffness in regions corresponding to the somatic representation of those states (diaphragm, psoas, posterior cervical chain), compared with matched controls.

This is measurable by ultrasound elastography independently of any subjective report, and the effect should be graded by trauma severity.

5.2 Bridge 2: Myofascial Release = Barrier Lowering

Physical claim (Schleip, 2003): Manual and movement-based interventions that target the fascial network produce measurable reductions in fascial stiffness and corresponding changes in interoceptive sensitivity and emotional availability.

Formal correspondence: These interventions reduce $|W_{ij}|$. They do not necessarily produce a state transition; they reshape the energy landscape to make transitions more accessible. If initial barrier is $W = -12$ and intervention reduces it to $W = -6$, QUANT-EXP-1 results (Johnson, 2026b) suggest that classical thermal dynamics can now cross what previously required quantum assistance.

Testable prediction (P2): The probability of emotional state transition following myofascial release should increase monotonically with the degree of reduction in fascial stiffness. This is testable by measuring both pre/post fascial stiffness (elastography) and pre/post emotional state (validated affect measures + HRV) in a within-subjects design across a series of somatic therapy sessions.

Testable prediction (P3): The phenomenological *character* of the transition should differ predictably: sessions that lower the barrier significantly should produce gradual, integrative shifts; sessions that trigger a crossing of a high barrier (large, rapid state transition) should produce different qualitative reports. The model predicts this without any additional assumptions.

5.3 Bridge 3: Therapist-Client Entrainment = Co-Identification

Physical claim (McCraty & Childre, 2010): In effective therapeutic contact, measurable physiological entrainment occurs between therapist and client — cardiac coherence synchronisation, mutual modulation of HRV spectra, and (in contact work) fascial tension synchronisation. This is not inferred; it is measured by simultaneous ECG and, in some studies, by direct force measurement.

Formal correspondence: This is the physical mechanism of **co-identification** (Johnson, 2026a) — the process by which the observer's soma-field is modified by contact with another's soma-field. The mathematical treatment describes this as a tensor product coupling; the physical implementation is fascial and electromagnetic entrainment. The therapist does not merely witness the client's state; the therapist's attractor landscape is temporarily modified by coupling to the client's, and this modification is the mechanism of therapeutic resonance.

Testable prediction (P4): The degree of measurable physiological entrainment (HRV coherence synchronisation) between therapist and client should predict therapeutic outcome — reduction in client fascial stiffness and shift in validated affect measures — independently of the specific technique used. Sessions with high physiological coherence should outperform sessions with low coherence, across modality.

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6 Testable Predictions

The bridges in §5 generate six primary empirical predictions, ordered from most to least accessible with current instrumentation:

#	Prediction	Method	Population
P1	CPTSD/complex-trauma populations show elevated fascial stiffness in diaphragm, psoas, posterior cervical chain vs matched controls	Shear-wave ultrasound elastography	CPTSD vs. controls ($n \geq 40$ per group)
P2	Somatic intervention reduces fascial stiffness; degree of reduction predicts probability of self-reported emotional state shift	Elastography pre/post + validated affect measures	Somatic therapy clients (within-subjects)

#	Prediction	Method	Population
P3	Barrier-lowering sessions (gradual stiffness reduction) produce qualitatively different transition phenomenology from barrier-crossing sessions (acute large shifts)	Mixed methods: elastography + structured interview	Rolfing or myofascial release series
P4	Therapist-client HRV coherence predicts session outcome independently of technique	Simultaneous ECG coherence + validated outcomes	Therapist-client dyads, multiple modalities
P5	Biophotonic emission from CPTSD populations differs from controls at characteristic emission bands (500–800 nm)	Ultra-weak photon measurement (photomultiplier)	CPTSD vs. controls

#	Prediction	Method	Population
P6	Transitions from Fear-dominant to Awe-dominant states (as defined by QUANT-EXP-1 attractor labels) correlate with measurable HRV spectral shift from LF-dominant to HF-dominant	HRV spectral analysis + soma-field state labelling instrument	Clinical transition cases

Predictions P1–P4 are testable with instrumentation available in clinical research centres now. P5 requires specialised biophoton detection (available in approximately a dozen research centres worldwide). P6 requires the prior development of a validated soma-field state classification instrument — a prerequisite for large-scale empirical work that is not yet available and is noted as the primary methodological gap in this programme.

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7 Conclusion

The Soma-Field model describes a field of emotional dynamics that is formally equivalent to a quantum field on an attractor manifold. This paper has argued that the physical substrate of that field consists of three interlocking systems:

1. The **biotensegrity network** (fascia, connective tissue, interstitium under prestress) that provides the continuous mechanical medium through which the somatic wave $\mathbf{E}_{\text{body}}$ propagates globally and rapidly (Ingber, 1997, 2003; Levin, 2002).
2. The **fascial interoceptive pathway** (type IV free nerve endings $\rightarrow$ lamina I $\rightarrow$ thalamus $\rightarrow$ insula) that constitutes the body-to-brain projection of somatic state (Craig, 2003; Schleip, 2003), and whose chronic remodelling under trauma — fascial armoring — is the physical implementation of the deep attractor basin.
3. The **bioelectric and biophotonic field** generated by the liquid crystalline living matrix and the cardiac electromagnetic environment (Ho, 1998; McCraty & Childre, 2010; Popp, 2003), which constitutes the best current physical candidate for the soma-field correlate itself.

The most clinically significant result of this identification is Bridge 1: the quantitative correspondence between fascial stiffness and attractor depth. This makes concrete a claim that somatic therapists have held for decades — that trauma is held in the body, not only in the mind — and extends it: the depth at which trauma is held is measurable by elastography, and the degree to which physical intervention changes that depth is also measurable. The soma-field model predicts that large barriers require quantum-assist crossing; the fascial model predicts that those same barriers are associated with measurable tissue-level changes. The two predictions are about the same phenomenon at two levels of description.

Bridge 3 — therapist-client entrainment as co-identification — connects this to the broader programme (Johnson, 2026a). The therapist's role is not neutral observation but active field coupling. The mathematics of co-identification (Johnson, 2026a) now has a proposed physical mechanism: fascial and electromagnetic entrainment, measurable, manipulable, and predictive of outcome.

This paper opens a research programme. The six predictions in §6 define the empirical agenda. The formal soma-field model provides the theoretical frame. The three bodies of literature reviewed here provide the biological grounding. Together they constitute a foundation for a genuinely interdisciplinary field — one that does not require the reader to choose between the body and the mathematics, because the mathematics is about the body.

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